



### 11.2.6 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each key feature.



I need help.



I got it.



I could help someone.

|                                                                      |                                                                                     |                                                                                     |                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Finding domain and range of an exponential function                  |    |    |    |
| Finding the constant percent rate of change                          |    |    |    |
| Describing the end behavior of an exponential function               |    |    |    |
| Finding the $y$ -intercept                                           |    |    |    |
| Finding the intervals where the function is positive or negative     |    |    |    |
| Finding the intervals where the function is increasing or decreasing |   |  |   |
| Identifying the horizontal asymptote                                 |  |  |  |

## Unit 11, Lesson 3: Graphing Exponential Functions Given an Equation



### Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



### Learning Targets

- ☐ I can graph an exponential function given the equation.
- ☐ I can classify a function as representing exponential growth or decay.



### 11.3.1 Warm-Up: Registered Nurse

1. Keith Koga is a registered nurse. He said he's had a lot of mentors, some of which included nurses he worked with. The culture of the workplace was one where everyone supported each other and knew they could go to each other for help.
  - a. Who is someone you consider a mentor in your life?
  - b. When learning something new, why is it helpful to ask someone (a peer, a teacher, a mentor, etc.) when you need help?



### 11.3.2 Guided Instruction: Graphing an Exponential Equation

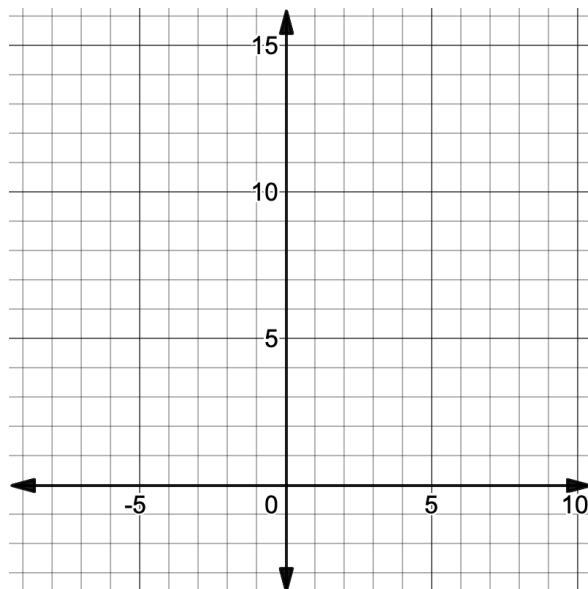
When graphing an exponential function, knowing the key features and using a table of values can be helpful.

1. Explain what key features can be determined from the equation  $C(x) = 5\left(\frac{6}{5}\right)^x$ .

| Key Feature                     | Explanation |
|---------------------------------|-------------|
| Growth or Decay                 |             |
| Domain and Range                |             |
| Increasing / Decreasing         |             |
| Constant Percent Rate of Change |             |
| End Behavior                    |             |
| Positive / Negative             |             |

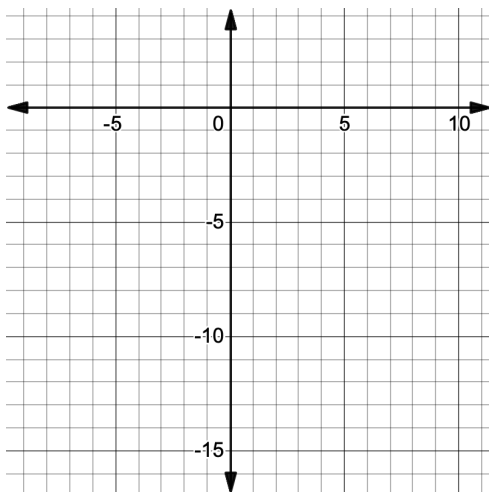
2. Complete the table for  $C(x) = 5\left(\frac{6}{5}\right)^x$ . Then, graph the function.

| $x$ | $C(x)$ |
|-----|--------|
| -2  |        |
| -1  |        |
| 0   |        |
| 1   |        |
| 2   |        |
| 3   |        |



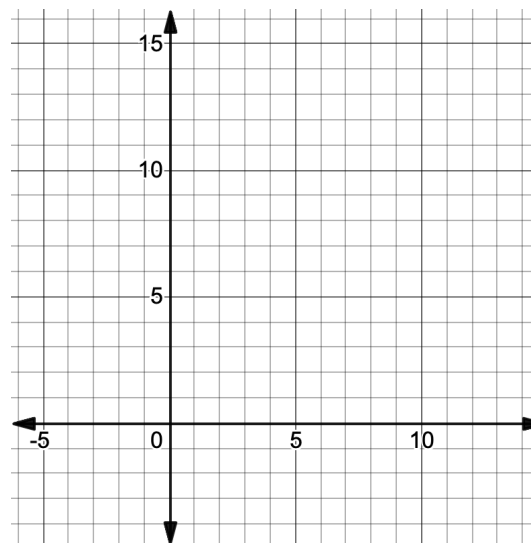
- a. Discuss with your partner how the graph would change if the  $a$ -value of 5 was changed to  $-5$ .

- b. Draw a sketch of  $C(x) = -5\left(\frac{6}{5}\right)^x$ .



- c. Discuss with your partner how the graph would change if the  $b$ -value was changed from  $\frac{6}{5}$  to  $\frac{5}{6}$ .

- d. Draw a sketch of  $C(x) = 5\left(\frac{5}{6}\right)^x$ .





### 11.3.3 Collaboration: Graphing Exponential Functions

1. Consider the function  $f(x) = 3(1 - 0.32)^x$ .
- a. Work with your partner to complete the statements about  $f(x) = 3(1 - 0.32)^x$ .

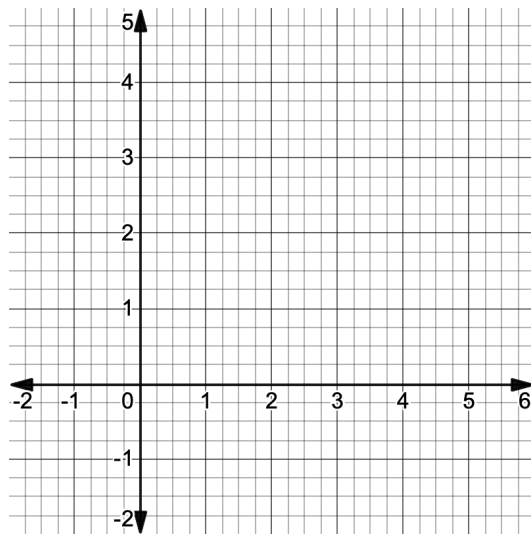
Since  $a$  is ☐ positive  
☐ negative and  $b$  is ☐ between 0 and 1,  
☐ greater than 1,

the function is ☐ increasing.  
☐ decreasing. The left end behavior of

the graph is ☐  $y \rightarrow 0$   
☐  $y \rightarrow \infty$  and the right end behavior is

☐  $y \rightarrow 0$ .  
☐  $y \rightarrow -\infty$ . The  $y$ -intercept of the function is ☐ 0.32.  
☐ 0.68.  
☐ 3.

- b. Sketch the graph of  $f(x) = 3(1 - 0.32)^x$ .



2. Consider the function  $f(x) = -(1 + 0.4)^x$ .

- a. Work with your partner to complete the statements for  $f(x) = -(1 + 0.4)^x$ .

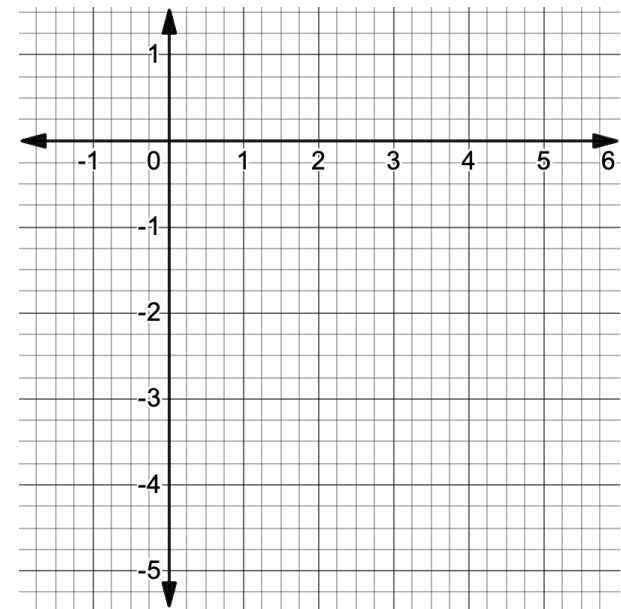
Since  $a$  is ☐ positive  
☐ negative and  $b$  is ☐ between 0 and 1,  
☐ greater than 1,

the function is ☐ increasing.  
☐ decreasing. The left end behavior of

the graph is ☐  $y \rightarrow 0$   
☐  $y \rightarrow -\infty$  and the right end behavior is

☐  $y \rightarrow 0$ .  
☐  $y \rightarrow -\infty$ . The  $y$ -intercept of the function is ☐ 0.4.  
☐ -1.  
☐ 1.04.

- b. Sketch the graph of  $f(x) = -(1 + 0.4)^x$ .





### 11.3.4 Activity: Card Sort

Work with your partner to complete the card sort. Fill in the table with the correct card labels that correspond to each graph.

| Card Sort |             |                |          |       |       |
|-----------|-------------|----------------|----------|-------|-------|
| Graph     | y-Intercept | $a =$<br>$b =$ | Function | Table | Table |
| 1.1       |             |                |          |       |       |
| 1.2       |             |                |          |       |       |
| 1.3       |             |                |          |       |       |
| 1.4       |             |                |          |       |       |
| 1.5       |             |                |          |       |       |
| 1.6       |             |                |          |       |       |



### 11.3.5 Your Turn!

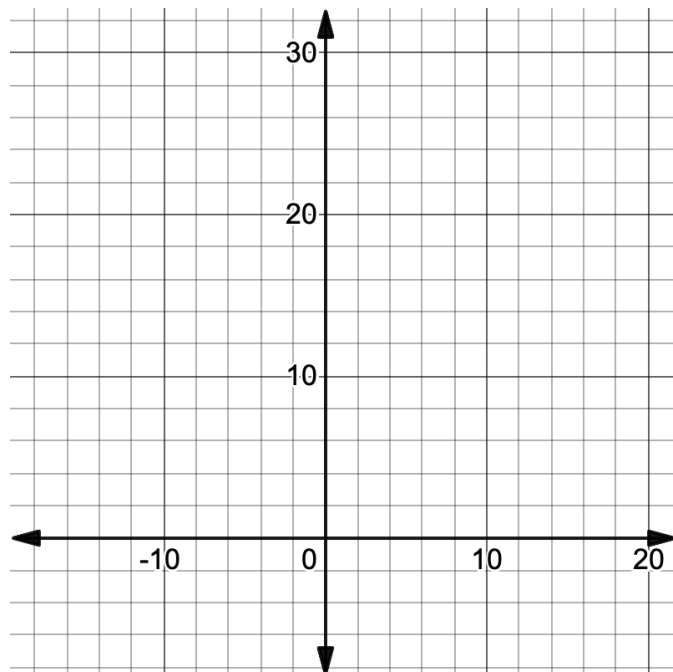
- Graph the exponential equation  $y = 45(1 - 0.05)^x$ . If necessary, complete the table of values. Then, determine all the key features.

| Graph                    |  | Table of Values                 |     |
|--------------------------|--|---------------------------------|-----|
|                          |  | $x$                             | $y$ |
|                          |  | -2                              |     |
|                          |  | -1                              |     |
|                          |  | 0                               |     |
|                          |  | 1                               |     |
|                          |  | 2                               |     |
| Key Features             |  |                                 |     |
| Domain                   |  | Range                           |     |
| y-intercept              |  | Constant Percent Rate of Change |     |
| Increasing or Decreasing |  | Positive                        |     |
|                          |  | Negative                        |     |
| Left End Behavior        |  | Right End Behavior              |     |



### 11.3.6 Wrap-Up: Ticket Out the Door

1. Graph the function  $y = 9\left(\frac{5}{3}\right)^x$ .



## Unit 11, Lesson 4: Graphing Exponential Functions Given a Written Description and Equation



### Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



### Learning Targets

- ☐ I can graph an exponential function given a written description and equation.
- ☐ I can determine the key features of an exponential function.
- ☐ I can classify a function as representing exponential growth or decay.